



Ductile Nb alloys with reduced density and cost

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ABSTRACT

Several Nb-based alloys containing Ti, Mo and W were produced and their microstructure, phase composition and mechanical properties in the temperature range of 23–1200 °C were reported and compared with a high-strength commercial Nb alloy C-3009, which contains Hf and W alloying elements. Replacement of Hf with Ti and W with Mo led to alloys with considerably reduced density and cost, as well as with reduced micro-segregation of the alloying elements. The strengthening effect from W was higher than from Mo at temperatures above ~600 °C and more amount of Mo was required to compensate the strength decrease. The strengthening effects from Hf and Ti were comparable at temperatures up to 1000 °C; however, at 1200 °C the effect from Hf was superior. It was concluded that the Nb–Mo–Ti system can be used as a baseline for the development of high strength, high-temperature alloys using alloying with other elements.

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1. Introduction

Nb-based refractory alloys currently used in some high-temperature engine applications contain expensive and dense alloying elements [1]. For example, C-103, which is one of the most commonly used medium-strength Nb alloys, contains (by wt.%) 10 Hf, 0.5 Ta, 0.5 W, 0.7 Zr and 1 Ti; and a high strength C-3009 contains 30 Hf and 9 W. Hf, Ta and Zr are expensive, costing approximately \$1200, \$290 and \$150 per kilogram, respectively, and Hf, W and Ta have high density of, respectively, 13.21, 16.65 and 19.25 g/cm³ (Table 1). Moreover, these alloys have poor oxidation resistance above 600 °C. It would therefore be beneficial to develop less expensive and lower density Nb alloys, which would have structural properties, such as high temperature strength and oxidation resistance, equivalent or better than those of the currently used commercial Nb alloys. There has been extensive research on Nb alloys containing Si, whose main goal is to improve both high temperature strength and oxidation resistance [2–6]; however, Nb–Si alloys are generally brittle at temperatures ≤ 1000 °C and they have not found practical use yet. Refractory complex concentrated alloys (RCCAs) is another promising direction of research resulting in developing several multi-principal element alloys (MPEAs) with attractive high temperature mechanical properties and improved oxidation resistance [7]. The main challenge is the

lack of experimental data, as well as valid theoretical models, on thermodynamic, mechanical and physical properties of these complex MPEA systems. RCCAs experience slow kinetics of phase transformations and may contain metastable phases. Thus extensive studies are required to identify equilibrium phase diagrams, kinetics of phase transformations and phase stability in these alloys in the time-temperature-stress conditions of practical interest [8,9].

In this work, we used conventional methods for alloy development and produced several Nb-based alloys, which contain Mo, W and Ti. A high-strength commercial Nb alloy C-3009 was used as a baseline material and the new alloys were produced by replacing heavy W and Hf with lower density Mo and Ti, respectively. Mo belongs to the same group (VI) of the periodic table as W and Ti to the same group (IV) as Hf. In addition to decreasing density, Ti is known to noticeably improve oxidation resistance of Nb [12,13]. However, Mo and Ti have lower melting temperatures than W and Hf. In addition, Mo has smaller elastic moduli than W but Ti has higher elastic moduli than Hf. Because the strength of the alloys generally increases with increasing in melting temperature and elastic moduli, replacement of W with Mo may reduce high temperature strength relative to C-3009, while the effect from replacement of Hf with Ti is not straightforward. Thus, the effect of the concentration of the alloying elements on high temperature properties has also been studied in this work.

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Table 1

Density and approximate cost of elements present in Nb alloys. Physical properties such as melting temperature, T_m , atom radius, r , Young's modulus, E , shear modulus, G , and bulk modulus, B , are also shown [10,11].

Element	Nb	Hf	Mo	Ta	Ti	W	Zr	Y
Density (g/cm ³)	8.57	12.31	10.22	16.65	4.51	19.25	6.51	4.47
Price (\$/kg) ^a	165	1200	66	290	66	44	150	500
T_m , K	2750	2506	2896	3290	1941	3695	2127.7	1795
r , pm	143	158	136	143.0	146.2	137	160	180
E , GPa	105	78	329	186	116	411	68	63.5
G , GPa	38	30	120	69	44	161	33	25.6
B , GPa	170	110	230	200	110	310	89	41

^a Price of metals is approximate and can vary with time.

2. Experimental procedures

A C-3009 alloy was received from ATI Wah-Chang, Inc. in the form of an as-cast ingot. Nb–W–Ti (W1), Nb–Mo–Ti (M1, M2, M3 and M4) and Nb–Mo–W–Ti (MW1) alloys were prepared from high purity (99.9% or higher) metals by vacuum arc melting. The elemental compositions of the alloys were determined using X-ray energy dispersive spectroscopy and are shown in Table 2. The alloy density was measured using a helium pycnometer, with the accuracy of ± 0.005 g/cm³. Vickers microhardness was determined as an average value of 10 measurements using a diamond pyramid applied to a polished surface of the alloy sample at 500 g load for 10 s. X-ray diffraction was conducted using Cu $K_{\alpha 1}$ radiation in the 2θ range of 10–120°. Microstructure and chemical composition were studied using a scanning electron microscope (SEM) equipped with backscatter electron and energy dispersive spectroscopy detectors.

Compression mechanical tests were conducted at 23 °C, 600 °C, 800 °C, 1000 °C and 1200 °C at a constant ram speed of 0.008 mm/s that corresponded to the initial strain rate of 0.001 s^{−1}. Compression specimens had a squared cross-section of 5 mm × 5 mm and the height of 8 mm. Special grinding tools were used to make the samples with parallel load surfaces and right-angle edge surfaces. Room temperature tests were conducted in air and high-temperature tests were conducted in vacuum of $\sim 2 \cdot 10^{-3}$ Pa. The temperature setting for high-temperature compression testing consisted of heating at 50 °C/min to a test temperature, holding at the temperature for 15 min, deformation with 50–55% height reduction (or until fracture, if it occurred first) and cooling at the initial cooling rate of 100 °C/min or the rate of furnace cooling, whichever was slower. To calculate a true sample displacement during loading, the following procedure was used. At a given testing temperature, the response of the loading system (machine frame, load cell, dies, etc.), without a sample, was first measured by registering displacement Δl_m as a function of the applied load P . Then a sample was placed between the dies and the displacement Δl_{sys} for the system with the sample was registered as a function of the applied load P at the same temperature and ram speed. After that, at every registered P_i values, the sample upset values $\Delta l_i(P_i)$ were calculated as $\Delta l(P_i) = \Delta l_{sys}(P_i) - \Delta l_m(P_i)$. The true strain was finally calculated as $\ln(1 - \Delta l/l_0)$ and true stress was calculated as $P(1 - \Delta l/l_0)/A_0$, where l_0 and A_0 are the initial

Table 2

Density (ρ), and chemical composition (in at.%), of the studied alloys.

Alloy	ρ (g/cm ³)	Nb	Mo	Ti	Fe	W	Hf
C-3009	10.3	71.7				5.9	22.4
W1	8.60	73.5		18.1		8.4	
M1	7.96	73.0	9.2	17.8			
M2	7.99	70.0	11.9	18.1			
M3	8.03	67.8	14.1	18.1			
M4	7.70	34.7	32.8	32.5			
MW1	8.94	32.9	17.0	34.4		15.7	

height and initial cross-section area of the sample. Room temperature deformation and elastic behavior of the samples was additionally recorded using Vic-Gage software by Correlated Solutions [14] and good agreement between these two methods was found. The room temperature elastic modulus values reported in this work were measured using Vic-Gage software.

3. Results and discussion

3.1. Microstructure and alloy density

According to X-ray diffraction, all the studied alloys are single-phase BCC structures (Fig. 1). The lattice parameters a_{mes} measured for the studied alloys are given in Table 3. It is found that a_{mes} is composition-dependent and has direct correlation with the atomic radii and concentrations of the alloying elements. For example, replacing larger Hf (alloy C-3009) with smaller Ti (alloy W1) decreases a_{mes} from 334.2 pm to 327.3 pm, and increasing the concentration of Mo, which is smaller than Nb and Ti, decreases a_{mes} from 329.5 pm to 326.5 pm in alloys M1–M3. The measured a_{mes} values are compared with the respective a_{ROM} values calculated using the rule of mixtures (ROM) of the alloying elements:

$$a_{ROM} = \sum c_i a_i = \frac{4}{\sqrt{3}} \sum c_i r_i \quad (1)$$

In Equation (1), c_i and r_i are the atomic fraction and atomic radius of an alloying element i taken from Tables 2 and 1, respectively, and $a_i = \frac{4}{\sqrt{3}} r_i$ is the BCC lattice parameter of element i [15,16]. The calculated a_{ROM} values are given in Table 3. It can be seen that a_{mes} are from 0.13% (for M1) to 1.62% (MW1) higher than a_{ROM} , with a larger difference for more concentrated alloys, indicating bond shortening in the alloys relative to pure metals.

The C-3009, W1 and M1 alloys have coarse grain structures, with the average grain size over 1.5 mm (Fig. 2a–c). A dendritic structure is clearly seen in the W-containing C-3009 and W1 (Fig. 2a and b). The light regions are rich in W and darker regions are rich in Hf (in C-3009) or Ti (in W1). Large difference in the atomic numbers of W and Ti provided stronger contrast between dendritic and interdendritic regions in W1 when compared to C-3009, in which W and Hf had smaller difference in the atomic numbers. The dendritic structure is absent in M1, which contains Mo instead of W. The chemical micro-segregation of the alloying elements between dendrites and interdendritic regions may change the local lattice parameter of the BCC crystal structure of the alloys. Therefore, the diffraction peak positions represent averaged distances between the respective crystal planes, because X-ray spectrums are taken from rather large surface areas, which contain many grains. In the Mo-containing alloys (M1 to M4) the average grain diameter decreases with an increase in the amount of Mo and it is 1.41 mm, 0.90 mm, 0.60 mm and 0.26 mm in M1, M2, M3 and M4 alloys, respectively (Fig. 2c–f). Traces of a dendritic structure are seen inside grains in M2 and M4 alloys. It should be noted that M4 is a near equimolar composition of Mo, Nb and Ti and, therefore, it cannot be strictly called Nb-based alloy. Partial replacement of Mo with W in M4 produced the MW1 alloy, which contains traces (less than 1% by volume) of a secondary phase, which is located at or near grain boundaries of the BCC matrix phase and has dark contrast on backscatter images (Fig. 2g). The average diameter of the matrix grains is 59 μ m. According to the EDS analysis, the composition of the matrix grains is close to the average alloy composition, while the secondary phase is rich in Ti (Table 4). EBSD phase analysis reveals BCC crystal structure of the Ti-rich secondary phase. All the alloys contain fine pores after casting. They are seen as black dots, mainly inside grains, on the backscatter images and

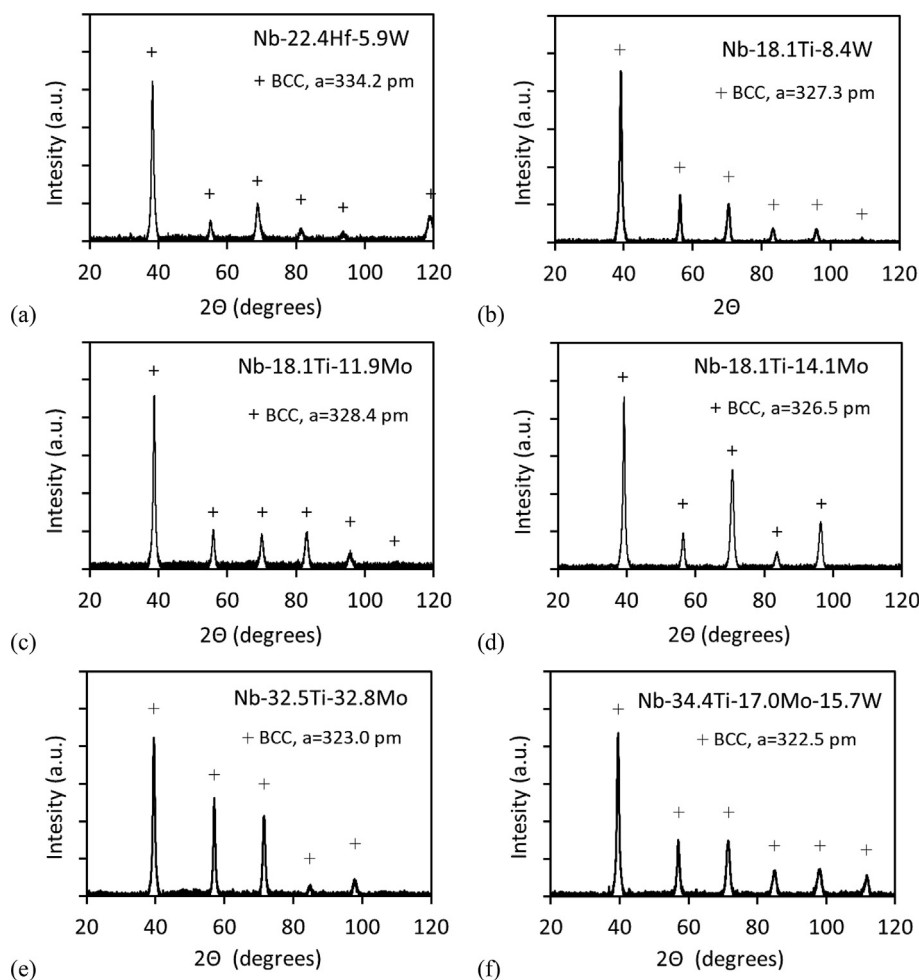


Fig. 1. X-ray diffraction patterns of (a) C-3009, (b) W1, (c) M2, (d) M3, (e) M4 and (f) MW1 alloys.

Table 3

Measured (a_{mes}) and calculated (a_{ROM} , Equation (1)) BCC lattice parameters for the studied alloys.

Alloy ID	C-3009	W1	M1	M2	M3	M4	MW1
a_{mes} , pm	334.2	327.3	329.5	328.4	326.5	323.0	322.5
a_{ROM} , pm	336.9	330.2	329.9	329.5	329.2	327.4	327.7

are the most populated in C-3009 (Fig. 2a). The volume fraction of these pores does not exceed 0.1%; thus only minor effect of these pores on compression ductility can be expected.

The density of the commercial alloy C-3009 measured with the use of a helium pycnometer is 10.3 g/cm^3 (Table 2). Replacement of Hf with Ti results in the W1 alloy with a reduced density of 8.6 g/cm^3 . Additional replacement of W with Mo decreases the density even further, so that M1 alloy density is 7.96 g/cm^3 . Densities of M2 and M3 alloys, which have increasing amounts of Mo, are 7.99 and 8.03 g/cm^3 , respectively. Densities of the multi-principal component alloys M4 and MW1 are 7.83 and 9.11 g/cm^3 . Thus, the replacements of W and Hf with Mo and Ti have clear beneficial effects on reducing the alloy density.

3.2. Mechanical properties

3.2.1. Room temperature properties

Room temperature mechanical properties were studied using

Vickers microhardness testing and compression testing. The measured microhardness (Hv) values are given in Table 5. Microhardness of C-3009 was 2.69 GPa . Replacement of Hf with Ti increased microhardness of W1 to 3.07 GPa . Microhardness of the developmental Nb–Mo–Ti alloys increased with increasing the amount of Mo from 2.17 GPa for M1 to 3.81 GPa for M4. Partial replacement of Mo in M4 with W increased microhardness of MW1 to 4.94 GPa . One can see that microhardness of all the developmental alloys, except M1, was higher than that of C-3009, with the largest increase ($\sim 84\%$) in MW1.

The room-temperature true stress vs. true strain curves for the studied alloys are shown in Fig. 4a and the measured yield stress ($\sigma_{0.2}$) and compression modulus (E) values are given in Table 5. At $T = 23^\circ\text{C}$, all the studied alloys showed continuous strengthening after plastic yielding. C-3009 had $\sigma_{0.2} = 663 \text{ MPa}$, while W1, in which Hf was replaced with Ti, had noticeably higher yield stress, $\sigma_{0.2} = 849 \text{ MPa}$. The Nb–Mo–Ti alloy M1 had the smallest $\sigma_{0.2}$ ($= 521 \text{ MPa}$) among the studied alloys, while $\sigma_{0.2}$ values of M2 and M3 (719 MPa and 780 MPa , respectively) were between those for C-3009 and W1. The multi-principal element alloys M4 and MW1 were considerably stronger showing $\sigma_{0.2}$ values of 1100 MPa and 1440 MPa , respectively. All the alloys, except M4 and MW1, showed homogeneous compression deformation, without any evidence of fracture. In the M4 and MW1 alloys, deformation localized inside macroscopic shear bands, which caused cracking and fracture along these bands at true compression strains of ~ 0.25 and 0.11 ,

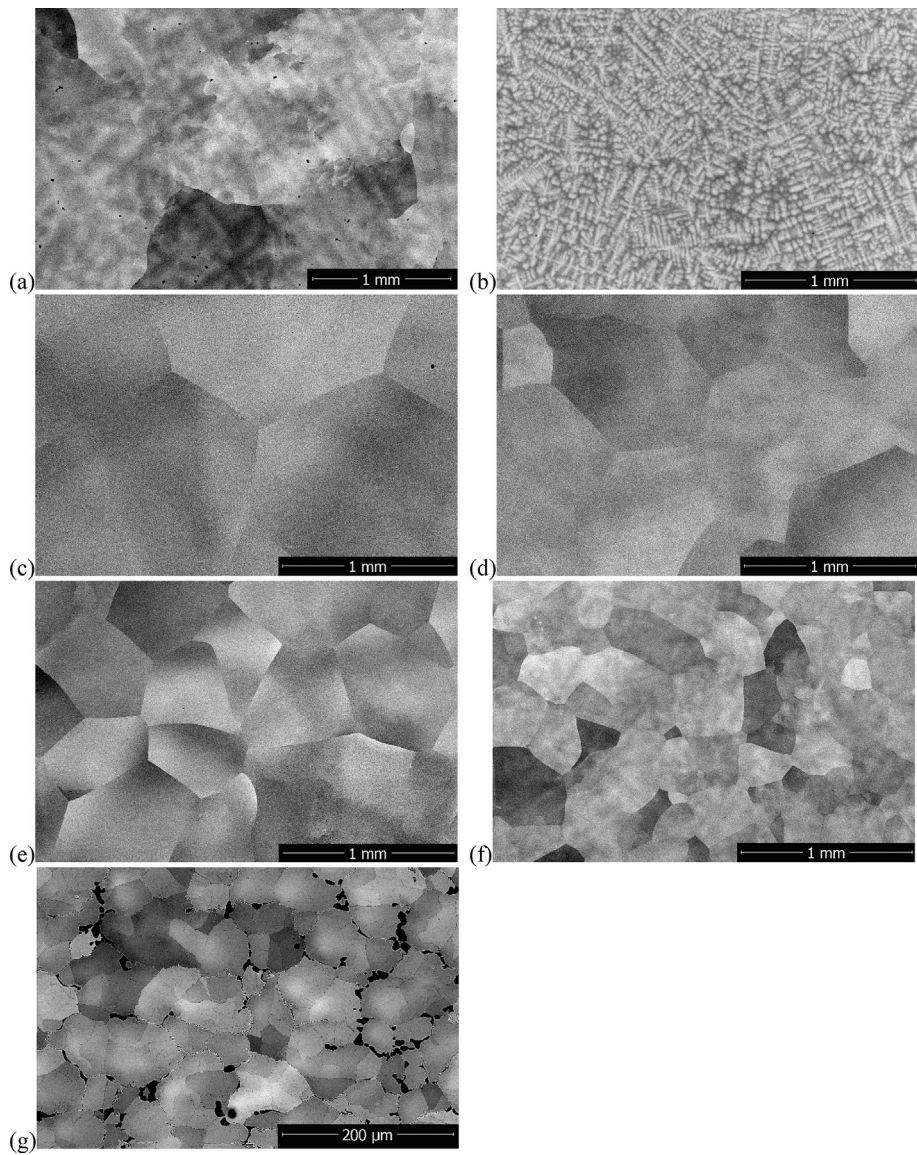


Fig. 2. SEM backscatter electron (BSE) images of the microstructure of (a) C-3009, (b) W1, (c) M1, (d) M2, (e) M3, (f) M4 and (g) MW1 alloys.

Table 4
Chemical composition (in at.%) of the matrix grains and second-phase particles in MW1 alloy.

Element	Nb	Mo	Ti	W
Matrix	34.4	17.4	32.6	15.6
Particles	8.9	1.2	89.3	0.6

Table 5
Room temperature mechanical properties of the studied alloys: Vickers microhardness (Hv), Young's modulus (*E*), yield stress ($\sigma_{0.2}$) and specific yield stress ($\sigma_{0.2}/\rho$). Elastic properties (Young's modulus E_{ROM} , shear modulus G_{ROM} and bulk modulus B_{ROM}) calculated using rule of mixtures, as well as the calculated number density of obstacles β , are also given here.

Alloy ID	C-3009	W1	M1	M2	M3	M4	MW1
Hv, GPa	2.69 ± 0.08	3.07 ± 0.08	2.17 ± 0.05	2.76 ± 0.06	2.81 ± 0.05	3.81 ± 0.06	4.94 ± 0.06
$\sigma_{0.2}$, MPa	663 ± 7	849 ± 8	521 ± 5	719 ± 7	780 ± 8	1100 ± 11	1440 ± 14
$\sigma_{0.2}/\rho$, MPa.cm ³ /g	65.6	98.7	65.5	90.0	97.1	142.9	161.1
<i>E</i> , GPa	110 ± 20	133 ± 20	125 ± 20	129 ± 20	138 ± 30	183 ± 30	193 ± 30
E_{ROM} , GPa	117	133	128	134	139	182	195
G_{ROM} , GPa	43	49	47	49	51	67	73
B_{ROM} , GPa	165	171	165	166	168	170	182
β , 10 ¹⁵ m ⁻²	2.78	3.67	1.53	2.68	2.97	3.46	4.95

respectively, after achieving peak true stress values of 1757 MPa and 1900 MPa (Fig. 4a). Due to their reduced density, the developmental alloys showed specific yield stress values superior to the specific yield stress of C-3009 (Table 5).

The measured compression modulus *E* of C-3009 was 110 ± 20 MPa (Table 5). The error in the measurements was rather

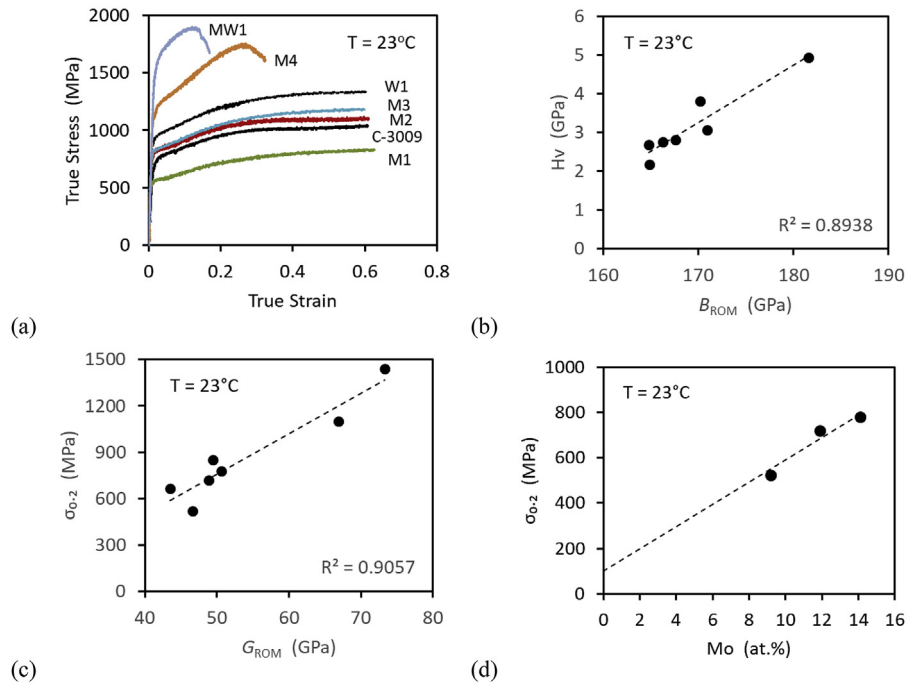


Fig. 3. Room temperature properties of the studied alloy: (a) true stress vs. true strain deformation curves; (b) Vickers microhardness vs. the bulk modulus; (c) yield stress vs. shear modulus; (d) yield stress of M1, M2 and M3 alloys vs. concentration of Mo.

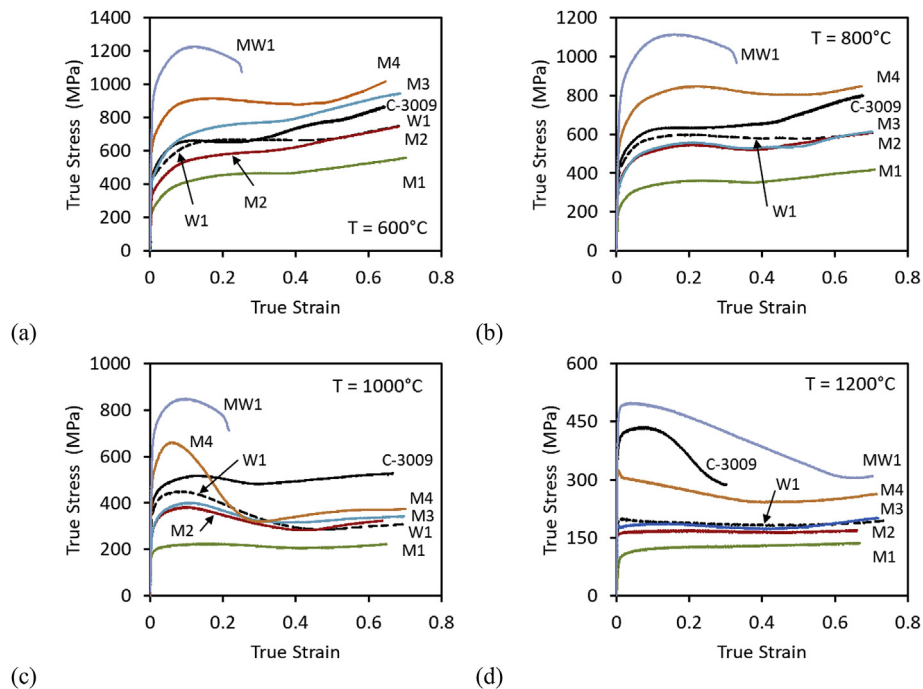


Fig. 4. True stress vs. true strain deformation curves of the studied refractory alloys at (a) 600 °C, (b) 800 °C, (c) 1000 °C and (d) 1200 °C.

high due to displacement signal noise. Replacement of Hf with Ti increased E to 133 GPa (alloy W1). The Nb-9.2Mo-17.8Ti alloy M1 had $E = 125$ GPa. Continuous increasing the amount of Mo led to an increase in E to 129 GPa, 138 GPa and 183 GPa in alloys M2, M3 and M4, respectively. The alloy MW1, which additionally contained W, was the stiffest, with $E = 193$ GPa. The measured E values were similar to the E_{ROM} values (Table 5) calculated using the rule of

mixtures of the Young's moduli of the pure elements from Table 1. These findings are in good agreement with earlier publications by Tian et al. [17] and Yao et al. [18] who used first-principle (FP) methods to calculate elastic moduli of many refractory BCC solid solution alloys and found that the calculated elastic constants are very close to those calculated using the rule of mixtures. Using the rule of mixtures, shear modulus G_{ROM} and bulk modulus B_{ROM} were

also calculated for the studied alloys and the calculated values are given in Table 5.

The measured microhardness and yield stress values were compared with the calculated elastic moduli of the respective alloys and useful relationships were found. In particular, the alloy microhardness was described as a linear function of the alloy bulk modulus and the yield stress roughly followed a linear dependence on the alloy shear modulus (Fig. 3b and c):

$$Hv = 0.149B_{ROM} - 22.0 \quad (2a)$$

$$\sigma_{0.2} = 0.026G_{ROM} - 541 \quad (2b)$$

Equation (2) are empirical equations, which are valid in a narrow range of B and G and may not have physical meaning, as they predict negative values of Hv and $\sigma_{0.2}$ at $B \leq 148$ GPa and $G \leq 20.8$ GPa, respectively.

The yield stress of a solid solution alloy, which contains obstacles for dislocation glide, with the obstacles' density β on the slip plane, can be described by the following equation [19]:

$$\sigma_{0.2} = \alpha b G(\beta)^{1/2} \quad (3)$$

Here α is a constant, which depends on the type of obstacles and can vary from ~0.3 for weak obstacles (e.g. solid solutions) to 3.0 for strong obstacles (e.g. precipitates), b is the Burgers' vector value and G is the alloy shear modulus. Using $\alpha = 1$, $b = \frac{\sqrt{3}}{2}a$, $a = a_{mes}$ from Table 3, $G = G_{ROM}$ and $\sigma_{0.2}$ from Table 5, the obstacle density β at the beginning of the plastic deformation was estimated for the studied alloys and the calculated values are shown in Table 5. The calculated β are in the range from $1.53 \times 10^{15} \text{ m}^{-2}$ for alloy M1 to $4.95 \times 10^{15} \text{ m}^{-2}$ for alloy MW1. These β values give the average distance between the obstacles of ~26 to 14 nm in the studied alloys. Such high number density of the obstacles in non-deformed single-phase alloys can only be related to solute atoms and their clusters. For example, the typical density of dislocation forest in annealed alloys is $\sim 1 \times 10^{11}$ to $3 \times 10^{12} \text{ m}^{-2}$, i.e. 3 to 4 orders of magnitude lower. The observed increase in β with increasing the concentration of the alloying elements in the same alloy families also supports the solid-solution strengthening mechanism in these alloys. For example, the observed increase in $\sigma_{0.2}$ with increasing the concentration of Mo in M1, M2 and M3 alloys can be described by a linear dependence (Fig. 3d):

$$\sigma_{0.2} = \sigma_0(1 + 0.49C_{Mo}) \quad (4)$$

Here C_{Mo} is Mo concentration (in at.%) and $\sigma_0 = 100$ MPa is the yield stress at $C_{Mo} = 0\%$.

3.2.2. Deformation behavior at elevated temperatures

The true stress vs. true strain deformation curves for the studied alloys at 600 °C, 800 °C, 1000 °C and 1200 °C are shown in Fig. 4. At these temperatures, W1, M1, M2, M3 and M4 did not fracture during deformation. MW1 had limited compressive ductility at 600 °C, 800 °C and 1000 °C, but it did not fracture at 1200 °C. C-3009 was ductile (did not fracture) at 600–1000 °C, but it fractured after true strain of 0.3 at 1200 °C.

At $T = 600$ °C and 800 °C, deformation of C-3009, M1, M2 and M3 occurred with continuous strain hardening, while steady-state flow at almost constant flow stress was observed in W1 and M4 (Fig. 4 a, b). MW1 showed continuous decrease in true stress after reaching a peak stress $\sigma_p = 1227$ MPa at $\varepsilon_p = 0.12$ ($T = 600$ °C) or $\sigma_p = 1115$ MPa at $\varepsilon_p = 0.16$ ($T = 800$ °C), which resulted in fracture at $\varepsilon_f = 0.3$ (600 °C) or 0.36 (800 °C). At 1000 °C and 1200 °C, all the

alloys showed strain hardening only at the beginning of deformation, up to ~0.1–0.18 true strain; then strain softening and/or steady state flow followed (Fig. 4 c, d). A considerable loss in compressive ductility was observed in C-3009 at 1200 °C (Fig. 4 d). Visual observation of the tested C-3009 samples showed that this loss in compressive ductility was due to strain localization in a single ~45°-inclined shear band and fracturing along this band. Such behavior indicates deformation instability of C-3009 at 1200 °C. Fine-grained structure and multiple cracks were observed inside this shear band. It is likely that formation of fine-grained structure within the shear band was responsible for strain localization and cracking. The fine-grain structure initiated extensive grain boundary sliding, which, however, was not yet accommodated by dislocations and diffusion inside grains, and this resulted in crack formation at triple junctions and along grain boundaries. By growing and merging, these micro-cracks led to macroscopic fracture. Obviously, the strain hardening capability of the alloy is not sufficient to initiate plastic deformation on 5 independent slip systems needed for homogeneous shape change.

During heating from 23 °C to 600 °C, all alloys showed considerable drop in $\sigma_{0.2}$ (Tables 5 and 6). For example, $\sigma_{0.2}$ of C-3009 decreased from 663 MPa at 23 °C to 437 MPa at 600 °C. A further increase in T to 1200 °C caused only slight (by ~10%) decrease in $\sigma_{0.2}$ of this alloy, to 388 MPa (Table 6), indicating high yield stress stability of C-3009 against heating from 600 °C to 1200 °C. The yield stress of W1, M1, M2 and M3 alloys at 600 °C was twice smaller than at 23 °C. Such a rapid decrease in strength resulted in these 4 alloys to become weaker than C-3009 at $T \geq 600$ °C. The W1 alloy, in which Ti was used instead of Hf, had almost the same $\sigma_{0.2}$ as C-3009 at 600 °C but its strength decreased quicker with increasing T and at 1200 °C $\sigma_{0.2}$ of W1 was twice smaller than that of C-3009 (Table 6). This result indicates that although the replacement of Hf with Ti makes the alloy stronger at room temperature, probably due to increasing elastic moduli, the Ti-containing alloy becomes softer at high temperatures, probably because of reduced melting temperature (Table 1, Table 5).

The M1 alloy, which contains 17.8% Ti and 9.2% Mo, was the least strong alloy at all the studied temperatures. It had $\sigma_{0.2} = 170$ MPa and 97 MPa at 1000 °C and 1200 °C, respectively. The high-temperature strength increased almost linearly with increasing the amount of Mo in M1, M2 and M3 alloys, and M3, which has 14.1% Mo, had $\sigma_{0.2} = 270$ MPa and 183 MPa at 1000 °C and 1200 °C, respectively (Table 6). It is evident that solid solution strengthening is a dominant strengthening mechanism in these alloys at all the studied temperatures.

The multi-principal element alloys, M4 and MW1 were considerably stronger than the other developmental alloys at all the studied temperatures (Table 6). Moreover, $\sigma_{0.2}$ of M4 was higher than that of C-3009 at all temperatures but 1200 °C and $\sigma_{0.2}$ of MW1 was superior to C-3009 in the temperature range from 23 °C to 1200 °C. Unfortunately, M4 showed limited compressive ductility at 23 °C and MW1 showed limited compressive ductility at

Table 6

Yield stress ($\sigma_{0.2}$, in MPa) / specific yield stress ($\sigma_{0.2}/\rho$, in MPa cm³/g), values of the studied alloys at different temperatures.

Alloy	T = 23 °C	T = 600 °C	T = 800 °C	T = 1000 °C	T = 1200 °C
C-3009	663 / 65.6	437 / 43.3	424 / 42.0	397 / 39.3	388 / 38.4
W1	849 / 98.7	420 / 48.8	389 / 45.2	348 / 40.5	194 / 22.6
M1	521 / 65.5	230 / 28.9	207 / 26.0	170 / 21.4	97 / 12.2
M2	719 / 90.0	350 / 43.8	311 / 38.9	250 / 31.3	161 / 20.2
M3	780 / 97.1	390 / 48.6	329 / 41.0	270 / 33.6	183 / 22.8
M4	1100 / 142.9	614 / 79.7	536 / 69.6	504 / 65.5	324 / 42.1
MW1	1440 / 161.1	883 / 98.8	747 / 83.6	635 / 71.0	461 / 51.6

all temperatures except 1200 °C. It is, however, necessary to point out that the properties of all the alloys were studied in as-cast condition and their compressive ductility is expected to increase after refining the microstructure by thermomechanical processing.

3.2.3. Using atomistic simulations to predict strength of refractory alloys

Recently Rao et al. [20,21] combined an analytical Suzuki model [22,23] with atomistic simulations of solute – screw dislocation core interaction energies to calculate the yield stress and predict its temperature dependence in several refractory-based complex concentrated alloys with a single-phase BCC crystal structure. Atomistic simulations were used to determine the solute – screw dislocation interaction energies, whereas the Suzuki model briefly described in the following paragraph, was used to determine the temperature dependence of the yield behavior using atomistic information.

The Suzuki model assumes that in concentrated BCC alloys kink pairs nucleate easily on screw dislocations while the kink migration along the screw dislocation line controls the rate of dislocation motion. Suzuki derived the activation energy for kink migration as [20–23]:

$$\Delta H(\tau_k) = 3cE_w^2\kappa^2/(2\tau_k ab^2) - \tau_k^3 a^3 b^4 \Lambda_k^2 / (18\kappa^2 E_w^2 c) \quad (5)$$

Where c is the concentration of solutes, E_w is the solute-screw dislocation first and second neighbor interaction energy, b is the Burgers vector of the screw dislocation, $a \sim 0.942b$, Λ_k is the kink width, τ_k is the resolved shear stress for the kink migration and κ is derived from the following equation [22,23]:

$$\frac{1}{\sqrt{2\pi}} \int_{\kappa}^{\infty} \exp\left(-\frac{x^2}{2}\right) dx = \frac{b}{3Lc} \quad (6)$$

Where L is the half-spacing between hard dipole obstacles on the screw dislocation. Using Orowan equation and taking the mobile screw dislocation density as $(5\pi kT/\mu b \Delta V)^2$, where ΔV is the activation volume [22,23], the strain rate, $d\epsilon/dt$ can be written down as

$$d\epsilon/dt = \nu ab (5\pi kT/\mu b \Delta V)^2 \exp(-\Delta H(\tau_k)/kT) \quad (7)$$

Where ν is the Debye frequency, k is the Boltzmann constant and T is the temperature. The stress τ_k required to satisfy the applied strain rate in Equation (7) as a function of temperature plus the stress required to bow the screw dislocation between the hard dipole obstacles ($\sim \mu b/4L$), is then minimized with respect to L and the critical resolved shear stress to move the screw dislocation is determined as a function of temperature. The solute-screw dislocation interaction energy, E_w , is obtained from atomistic simulations as detailed elsewhere [20,21]. The contribution from various solutes are then summed up according to a multi-component hardening law, as described in Refs. [20,21].

Here we apply the approach by Rao et al. [20,21] to calculate $\sigma_{0.2}$ of the studied alloys at different temperatures in order to further estimate the level of prediction and limitations, as the Suzuki model does not take into account diffusion-controlling processes. A Taylor factor $M = 2.73$, corresponding to a pencil-glide model in a texture-less BCC polycrystal [24], was used to convert the calculated critical resolved shear stress τ_k to the compression yield stress $\sigma_{0.2}^{calc}$. Table 7 shows the calculated $\sigma_{0.2}^{calc}$ values for the studied alloys and the percent of deviation ($= \frac{\sigma_{0.2}^{calc} - \sigma_{0.2}^{mes}}{\sigma_{0.2}^{mes}} \cdot 100\%$) of the calculated values from the respective experimentally-measured $\sigma_{0.2}^{mes}$, given in Table 6. The substitutional solute – screw dislocation interaction

Table 7

The yield stress ($\sigma_{0.2}^{calc}$) calculated using Suzuki model for the studied alloys at different temperatures and its deviation (in %) from the respective experimental value ($\sigma_{0.2}^{mes}$) given in Table 6.

Alloy	T = 23 °C	T = 600 °C	T = 800 °C	T = 1000 °C	T = 1200 °C
W1	830 / 2.2	382 / 9.5	335 / 13.9	287 / 15.5	257 / 32.5
M1	651 / 25.0	291 / 26.5	253 / 22.2	216 / 27.1	195 / 101
M2	765 / 6.4	342 / -2.3	299 / -3.9	255 / 2.0	230 / 42.9
M3	841 / 7.8	377 / -3.3	330 / 0.3	280 / 3.7	253 / 38.3
M4	1645 / 4 9.5	725 / 18.0	616 / 14.9	538 / 16.7	477 / 47.2
MW1	1884 / 30.8	833 / -5.7	708 / -5.2	617 / -2.8	550 / 19.3

energies were obtained from atomistic simulations using simple Johnson-Zhou interatomic potentials [25,26]. Satisfactory agreement between calculated and experimental $\sigma_{0.2}$ values was found for W1, M2, M3 and MW1 alloys in the temperature range from 23 °C to 1000 °C. However, ~20–26% difference between $\sigma_{0.2}^{calc}$ and $\sigma_{0.2}^{mes}$ was noted for alloys M1 and M4. The noticeably higher $\sigma_{0.2}^{calc}$ values (relative to $\sigma_{0.2}^{mes}$) in M1 could be due to high Taylor factor ($=2.73$) used in the calculations. Indeed, $M = 2.73$ is generally valid for randomly oriented polycrystalline structures. On the other hand, the M1 alloy consisted of very large grains, and the small number of grains in the compression samples could effectively reduce the Taylor factor closer to that of softer orientations [27]. Indeed, the calculations for M1 would well agree with the experiment if $M = 2.25$ were used. The large deviation of the calculated $\sigma_{0.2}$ values from the experimental ones for the multi-principal element alloy M4 could be due to errors in the atomistic calculations of the solute – screw dislocation interaction energy parameters, e.g. due to neglecting far-neighbor interactions. First principles calculations need to be used for a more rigorous calculation of these quantities [28]. The experimental data in Table 6 seems to show a faster decrease of $\sigma_{0.2}$ between 1000 °C and 1200 °C, as compared to the Suzuki model results (Table 7). The large deviation between the experimental and calculated values at 1200 °C is likely due to the onset of diffusion-controlled processes, which the Suzuki model does not consider [20]. Thus the Suzuki model can satisfactorily predict yield stress of the Nb refractory alloys at temperatures ≤ 1000 °C, but it fails at higher temperatures, when the diffusion-controlled processes start to dominate.

3.2.4. Improving specific strength

Although the yield stress of C-3009 was superior to the developmental alloys W1, M1, M2 and M3 at $T \geq 600$ °C, its specific yield stress, due to higher density, lagged up to 1000 °C (Table 6, Fig. 5). For example, at room temperature, $\sigma_{0.2}/\rho$ of C-3009 and M1 are the same ($=65.5$ MPa cm³/g) and much smaller than those of W1, M2 and M3. At 600 °C, $\sigma_{0.2}/\rho$ of C-3009 is similar to M2 but smaller than that of M3 and W1. At 800 °C, $\sigma_{0.2}/\rho$ of C-3009 is almost similar to that of M3; but at $T \geq 1000$ °C, C-3009 became superior to these three Mo-containing alloys. At the same time, $\sigma_{0.2}/\rho$ of W1 is superior to M1, M2, M3 and C-3009 at temperatures up to 1000 °C. The specific yield stress values of M4 and MW1 are considerably higher than those of all other studied alloys, including C-3009. It is apparent that solution strengthening of Nb by Mo is noticeably weaker than by W. Alloying with higher amount of Mo is required to achieve the strengthening effect similar to W. At the same time, Ti provides stronger strengthening effect than Hf at room temperature; however, the advantage is lost above 800 °C.

4. Conclusions

Six new Nb-based refractory alloys containing W and Ti (W1) or Mo and Ti (M1, M2, M3 and M4) or Mo, Ti and W (MW1) were

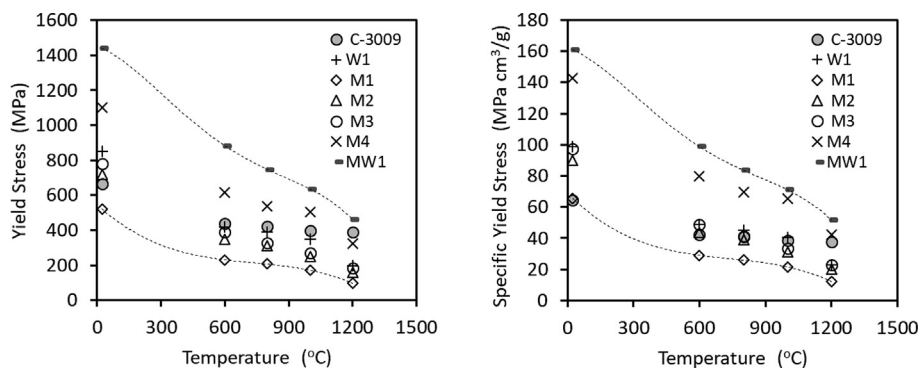


Fig. 5. Temperature dependence of (a) yield stress and (b) specific yield stress of the studied alloys. Dotted lines are trend-lines of the properties of alloys M1 and MW1, which showed, respectively, the lowest and highest yield stress values.

produced and their microstructure and compression properties in the temperature range of 23–1200 °C were reported and compared with commercial C-3009.

The produced alloys, except MW1, had single-phase BCC crystal structures. MW1 had small fraction of Ti-rich precipitates located at grain boundaries. While chemical micro-segregation was clearly observed in the cast C-3009, replacement of Hf with Ti and W with Mo led to more homogeneous distribution of the alloying elements in the developmental alloys. In Mo-containing alloys, the average grain size decreased with an increase in the amount of Mo.

The densities of the produced Nb alloys were considerably smaller than the density of C-3009 (10.3 g/cm³). Alloy M4 had the smallest density (=7.70 g/cm³) and MW1 had the largest density (=8.94 g/cm³) among the reported alloys.

W-containing alloys (C-3009, W1 and MW1) were stronger than Mo-containing alloys at T = 600–1200 °C. With increasing temperature in this temperature range, Mo-containing alloys lost their strength quicker than W-containing alloys.

Alloying with Hf had beneficial effect on high-temperature strength relative to alloying with Ti; however, the effect from Ti was stronger at room temperature. On the other hand, alloying with Ti reduced the alloy density and increased specific strength at higher temperatures.

At all the studied temperatures, the strength of Mo-containing alloys increased almost linearly with increasing Mo concentration evidencing solid-solution strengthening.

Atomistic simulation based on Suzuki's model can satisfactorily predict yield stress of the Nb refractory alloys at temperatures ≤1000 °C, but it fails at higher temperatures, when the diffusion-controlled processes start to dominate.

Due to reduced density, the room temperature specific strengths of the developmental alloys were superior to that of C-3009. Specific strengths of M4 and MW1 were superior to C-3009 and all other reported alloys at all the studied temperatures.

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